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Closed-form analysis of the global–local buckling behavior of sandwich columns with additively manufactured lattice cores

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Abstract This study provides a comprehensive analysis of the global (in-plane and out-of-plane) and local (intracell and wrinkling) buckling behavior of sandwich columns with monolithically designed aluminum facesheets and face-centered body-centered cubic (FBCC) lattice cores. Approximate and numerical methods are employed to evaluate the influence of geometric parameters on buckling performance. A novel closed-form, higher-order approach is developed, incorporating core transverse compressibility and a refined displacement field. The finite element method (FEM) is employed to verify the approximate results for sandwich columns under various boundary conditions, using 3D solid elements for the facesheets and beam elements for the lattice core. The results demonstrate strong agreement with the closed-form approximate predictions, capturing both global and local buckling modes while revealing that the boundary conditions significantly affect global buckling but have a rather small influence on the local buckling behavior. The proposed approach offers enhanced accuracy and convergence with numerical methods, providing an efficient framework to analyze wrinkling failure modes in sandwich columns with lattice cores.

Keywords Sandwich · 3D Lattice core · Higher-order analysis · Global buckling · Wrinkling · Finite element analysis

1 Introduction

The flexibility of additive manufacturing techniques enables the design of novel lightweight materials in beams, plates, and shells, leading to new research on their mechanical challenges. Among these lightweight structures, sandwich structures with lattice cores have emerged as a versatile choice, offering not only excellent mechanical performance but also adaptability to various loading scenarios. Unlike traditional honeycomb or foam cores, lattice cores provide enhanced design freedom, enabling the optimization of properties such as stiffness, strength, and energy absorption through customized unit cell geometries. This adaptability makes lattice-based sandwich structures particularly appealing for applications where multifunctionality and structural efficiency are paramount. Selective Laser Melting (SLM), a cutting-edge additive manufacturing (AM) process, facilitates the production of intricate lattice geometries with review high precision and design flexibility. By manufacturing the facesheets and lattice core monolithically, SLM overcomes traditional limitations associated with bonding interfaces, such as adhesive failure, ensuring a homogeneous mechanical response. This capability is instrumental in exploring advanced design possibilities for lightweight, high-strength sandwich structures.

It is essential to refer to the studies conducted thus far on the stability issues of sandwich structures designed using additive manufacturing, as they provide the groundwork for this research. These studies demonstrate

that the combined use of analytical and numerical methods enables a detailed examination of the relationship between sandwich geometry and structural stability, offering insights into optimizing these structures for enhanced buckling resistance. By addressing local and global buckling phenomena in various geometric configurations, the literature has significantly advanced the design and analysis of lattice-core sandwich structures, leading to comprehensive and effective solutions. Early studies on carbon-fiber reinforced plastics (CFRP) pyramidal truss cores identified critical failure mechanisms and provided foundational insights into global stability [1,2]. Similarly, investigations of multilayer pyramid lattice cores advanced our understanding of buckling mechanisms under axial loads, highlighting their application potential in high-performance systems [3–5]. Investigations of the lattice cores of BCC (body-centered cubic), BCCZ (body-centered cubic with Z-directional reinforcement), FCC (face-centered cubic), and F₂BCC (face-to-body-centered cubic) lattice cores highlighted the critical role of the core geometry in influencing stability under compression [6]. Comprehensive analyses of sandwich panels established a strong correlation between core geometry, material properties, and overall stability, providing robust frameworks to improve buckling resistance [7]. Three-dimensional solutions [8] on the effects of transverse shear and web shear [9] on the sandwich columns provided critical insights into their influence on stability. Advanced studies on additively manufactured lattice-core sandwich structures further highlighted the interaction between core design and buckling transitions [10]. Improvements in lattice-core sandwich designs have emphasized the optimization of structural stability. The development of meta-lattice sandwich panels introduced designs capable of mitigating vibrations while maintaining structural stability [11,12]. Studies on BCC and rhombic dodecahedron (DOD) lattice sandwich cores demonstrated the effectiveness of combining analytical and numerical frameworks to evaluate local and global buckling resistance under diverse loading scenarios [13,14]. Graded lattice beams were optimized for enhanced strength and strut buckling resistance, revealing the critical role of geometric parameters in achieving stability [15,16]. A semi-analytical method was also developed to model the buckling of the functionally gradient material sandwich beam (FGM), which provides improved accuracy and computational efficiency [17]. In addition to parametric buckling analysis of a cylindrical panel [18,19], stochastic buckling analysis of a sandwich plate using higher-order modes highlighted the sensitivity of lattice sandwich structures to imperfections and variability [20]. Blast-resistant lattice sandwich cores were validated for their wrinkling and buckling resistance under extreme loading conditions, offering innovative solutions for demanding environments [21,22]. A novel model has also been proposed to improve the accuracy of buckling and wrinkling load predictions in sandwich beam-columns, achieving these advancements within a unified analytical framework [23]. The reviewed studies reveal that a profound understanding of geometry-driven mechanisms, material response, and instability behavior is vital for the effective design and optimization of lattice-core sandwich structures.

This paper presents a closed-form approximate investigation into the global (in-plane and out-of-plane) and local (intracell and wrinkling) buckling behavior of sandwich columns featuring aluminum facesheets and lattice cores. A novel closed-form approximate solution is developed that incorporates the core's transverse compressibility through a higher-order displacement field. To validate the approximate model, finite element method (FEM) simulations are conducted, and the buckling response of sandwich columns is examined under various boundary conditions. The proposed approximate approach is shown to provide highly accurate predictions for both global and local buckling behavior, while offering a substantially more efficient alternative to FEM simulations.

2 Materials and design

It is assumed that the facesheet and lattice core materials used in the sandwich structure design will be produced using AlSi10Mg powder with LSM. The dimensions of the sandwich structure and unit cell are presented below (Fig. 1). Because the sandwich is considered to be produced in an integral design, and the core and facesheet are composed of the same material (AlSi10Mg), there is no need for any adhesive.

To evaluate these mechanical properties accurately, the formulas shown in Table 1, provided by Xia et al. [24], are used. They developed a reduced-order mechanical model specifically for rod-like lattice structures, enabling the calculation of key core properties such as elastic modulus, shear modulus, and Poisson's ratio. They developed a reduced-order mechanical model specifically for rod-like lattice structures, enabling the calculation of key core properties such as elastic modulus, shear modulus, and Poisson's ratio. The accuracy of these properties was also validated through finite element simulations [25]. These properties are functions of the ratio of the strut diameter to the cell size (γ) and the elastic modulus of the base material (E_0).

Under uniaxial compressive loading, both global and local failure modes were observed in the sandwich column. The analysis considered in-plane and out-of-plane global buckling as well as intracell and wrinkling

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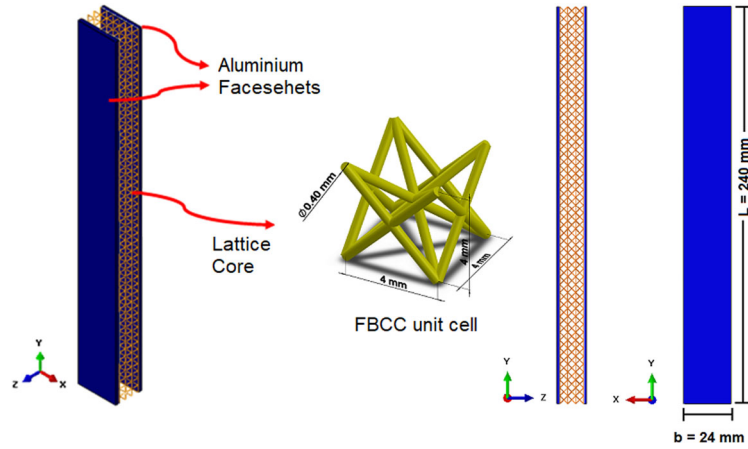


Fig. 1 Lattice sandwich column with FBCC lattice core and dimensions (the FBCC lattice cell is shown here as a representative cell type. All cells are $4 \times 4 \times 4$ mm in size and the strut diameter is 0.4 mm)

Table 1 Unit cell properties [24]

Name	Formula
FBCC	$E_x = E_y = E_0(1.661\gamma^{2.045}), \quad E_z = E_0(2.442\gamma^{2.953})$ $G_{xy} = E_0(0.6748\gamma^{2.045}), \quad G_{yz} = G_{zx} = E_0(1.275\gamma^{2.015})$ $v_{xy} = v_{yx} = 2.46\gamma^2 + 1.526\gamma - 0.5568$ $v_{xz} = v_{yz} = -2.551\gamma^2 - 1.223\gamma + 1.038$ $v_{zx} = v_{zy} = -1.27\gamma^2 - 0.01504\gamma + 0.6685$

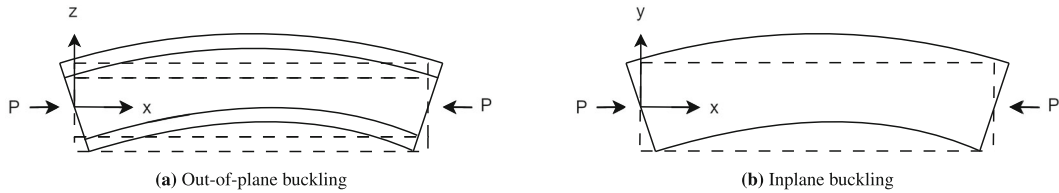


Fig. 2 Global buckling modes in a sandwich column

local buckling failure modes, which were examined separately using both analytical methods and FEM. The subsequent sections present a detailed exploration of these failure modes and the corresponding solutions. In the context of wrinkling analysis, it was found that existing solutions in the literature did not converge to FEM results within acceptable error margins. To address this limitation, the study introduces an original displacement formulation, as detailed in the section on wrinkling, which yields significantly improved results, thereby demonstrating the originality and contribution of this work.

3 Analytical methods for buckling prediction

3.1 Global buckling

Global buckling refers to the loss of overall structural stability of a sandwich column under axial compressive loading, resulting in large-scale deformation that affects the entire structure. It is divided into two kinds of buckling modes according to the buckling direction of the sandwich column plane.

1. **Out-of-Plane Buckling:** Out-of-plane buckling occurs when the sandwich column buckles perpendicular to its in-plane, specifically along the thickness (z).
2. **In-plane Global Buckling:** In-plane buckling refers to the deformation of a sandwich column within its plane, specifically along the width or length (xy -plane, as shown in Fig. 2).

Structures typically buckle out-of-plane along their major axis due to the weak bending stiffness of the related cross-section. However, in sandwich columns where the facesheet and core thickness vary, the weaker axis may shift beyond a certain threshold. The buckling direction is determined by the axis with the lower bending stiffness. Therefore, the bending stiffness of the sandwich column D_s , which will be used in the Euler-type critical buckling load formula P_E , must be evaluated for both the major and minor axes to identify the buckling direction. This sandwich bending stiffness D_s is determined separately for the out-of-plane modes, $D_{s,out}$, and in-plane modes, $D_{s,in}$, by the following formulas:

$$P_E = \frac{\pi^2 D_s}{(KL)^2}, \quad D_{s,out} = \frac{E_f b f d^2}{2}, \quad D_{s,in} = \frac{E_f f b^3}{6} + \frac{E_c^i b^3}{12} \quad (1)$$

Therein, E_f represents the facesheet modulus, E_c^i represents the core modulus, and i represents the lateral axis while K denotes the boundary condition coefficient. Out-of-plane and in-plane global buckling failure modes are influenced by loading conditions, boundary constraints, and the material's dimensional and mechanical properties. A sandwich structure tends to deform perpendicular to the plane, exhibiting lower bending and shear stiffness. Parametric analyses indicate that variations in the thickness of the facesheets and the core significantly influence the direction of buckling deformation, highlighting the relationship between structural parameters and buckling behavior.

The global buckling approaches given in Table 2, which are widely used in the literature, were adopted in this study. These approaches are fundamentally based on the Engesser approach (see Table 2), which refines Euler's buckling load formula by incorporating transverse shear effects, particularly for moderately thick columns. The key characteristics of the global buckling approaches utilized in this study are outlined as follows:

- **Engesser's Approach:** Considers the transverse shear strain γ , which is the change in the slope of the deflection curve due to the transverse shear force Q . The relationship is given by:

$$\gamma = \frac{\beta Q}{AG_{eq}} \quad (2a)$$

- **Haringx's Approach:** Includes the slope θ due to bending and extends Engesser's approach by defining the total slope of the deflected curve as:

$$\text{Total Slope} = \theta + \gamma \quad (2b)$$

- **Allen's Formulas:** Neglects core shear deformation, focusing on bending stiffness dominated by the face sheets. The thick face formula accounts for the inertia of thick faces and their axial stiffness through the quotient

$$\frac{I_f}{I_1} \quad (2c)$$

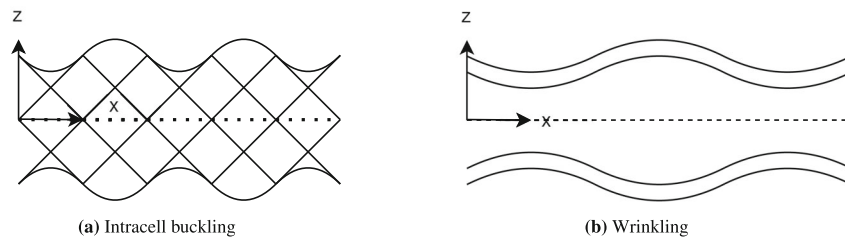
- **Bazant's and Cedolin's Method:** Refined shear correction by incorporating both core and face sheet effects, effectively addressing low core stiffness while balancing face sheet contributions. The equation is:

$$\beta = \frac{1}{G_c A_1 \left(\frac{1+I_f}{I_1} \right)} \quad (2d)$$

Allen's formula was chosen for FEM comparison because it distinguishes between thin and thick facesheets, incorporates the facesheet bending stiffness, and provides accurate and convergent results for both in-plane and out-of-plane buckling modes. Additionally, since the FBCC unit cell is considered as a soft core structure, Allen's formula is suitable as it was specifically developed for soft cores, disregarding the modulus of elasticity of the core in the bending stiffness determination.

Table 2 Global buckling approaches (for detailed information on the notation of approximations, see the references)

Author	Formula
Engesser [26]	$P_{cr} = \frac{P_E}{1 + \frac{\beta P_E}{AG_{eq}}}$
Haringx [27]	$P_{cr} = \frac{\sqrt{1 + \frac{4\beta P_E}{AG_{eq}}} - 1}{\frac{2\beta}{AG_{eq}}}$
Allen [28]	Thin Formula: $P_{cr} = \frac{AG_c P_E}{AG_c + P_E}$ Thick Formula: $P_{cr} = P_E \left[\frac{1 + \frac{P_E f}{P_c} - \frac{P_E f^2}{P_c P_E}}{1 + \frac{P_E}{P_c} - \frac{P_E f}{P_c}} \right]$
Bazant and Cedolin [29]	$P_{cr} = \frac{A_1 G_c P_E (I_1 + I_f)}{A_1 G_c I_1 + A_1 G_c I_f + I_1 P_E}$
Huang & Kardomateas [30]	$\frac{\beta}{AG_{eq}} = \frac{2b}{(EI)_{eq}^2} \left(\frac{a_f}{G_f} + \frac{a_c}{G_c} \right)$ $a_f = \frac{E_f^2}{4} \left[(f+c)^4 f - \frac{7}{15} (f+c)^5 - \frac{c^5}{5} + \frac{2}{3} (f+c)^2 c^3 \right]$ $a_c = E_f^2 f^2 c \left(\frac{f}{2} + c \right)^2 + \frac{2}{15} E_c^2 c^5 + \frac{2}{3} E_f E_c f \left(\frac{f}{2} + c \right) c^3$

**Fig. 3** Local buckling modes in sandwich columns with lattice cores

3.2 Local buckling

Local buckling is a structural instability phenomenon that occurs within specific regions of a sandwich structure, as opposed to global buckling, which involves deformation of the entire structure. Local buckling is characterized by localized deformations within the core or face sheets, which can significantly decrease the load carrying capacity of the structure and lead to premature failure. One of the most common forms of local buckling in sandwich structures is the buckling of the face sheets, which typically occurs as either intracell buckling or wrinkling. The key distinction between these two buckling modes lies in their respective wavelengths. Intracell buckling is associated with shorter wavelengths and often occurs when the face sheet is thin or the core features small, closely spaced cells, such as in lattice or honeycomb cores. Wrinkling, on the other hand, involves longer wavelengths and is more prevalent when the core is relatively compliant or thick (Fig. 3).

3.2.1 Intracell buckling

Intracell buckling is a form of local buckling that can occur in sandwich structures with cellular cores. Often referred to in the literature as “dimpling” or “intracell buckling”, intracell buckling describes the buckling of the face sheet within an individual lattice cell. Although the lattice structure may retain some load-bearing capacity after intracell buckling occurs, the resulting deformation in the face sheet can have undesirable effects, such as negatively impacting the structure’s aerodynamic properties. Therefore, it is essential to determine the load at which intracell buckling initiates. The earliest formula for intracell buckling, derived empirically from tests, was proposed by Norris and Kommers [31] specifically for honeycomb cores. Since then, there has been limited research into this particular buckling mode. The classical approach for predicting intracell buckling

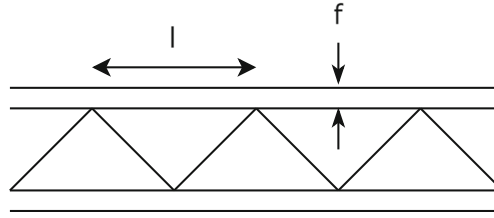


Fig. 4 Sandwich with corrugated core

remains largely empirical, with the fundamental assumption that each intracell plate (i.e., sections of the sandwich face supported along the edges of the honeycomb core cells) behaves similarly to a rectangular plate simply supported on all sides. Similar to Norris, Zenkert [32] also proposed a formula for a corrugated core to calculate the critical load using the buckling theory for an ordinary homogeneous plate. It results in:

$$P_{cr} = \frac{2k\pi^2 E^f}{12(1 - (v^f)^2)} \left(\frac{f}{l} \right)^2 \quad (8)$$

Therein, k is the buckling coefficient that depends on the size of the plate and the restraint, i.e., the edge conditions and the number of wavelengths in the buckling mode. The quantity E^f is the Young's modulus of the facesheet material, and v^f is the Poisson's ratio of the facesheet material. According to studies performed by Deshpande et al. [33] and Cote et al. [34], the assumption was introduced that the ends of the lattice structure are pin-jointed, which obtained good agreement with experiments. This leads to a value of $k=1$ for the buckling coefficient. The quantity l is the length between two adjacent struts, as shown in Fig. 4.

Existing approaches, such as the empirical formulations by Norris and Kommers for honeycomb cores and Zenkert's analytical solutions for corrugated cores, provide foundational insights into intracell buckling. The assumption of pin-jointed lattice structures by Deshpande and Cote aligns well with experimental observations. Among these, Zenkert's formula is selected in this study due to its compatibility with the periodic and porous nature of lattice cores, offering a reliable estimation for critical intracell buckling loads.

3.2.2 Wrinkling

Wrinkling, a common local stability issue in sandwich structures, refers to the formation of short-wavelength buckles in the face layers. This phenomenon can significantly decrease the stiffness and load-bearing capacity of the structure. In the literature, the wrinkling problem is typically categorized into two distinct modes: the symmetrical mode and the antisymmetrical mode. This failure mode arises from the interaction between the sandwich core and the face layers, which are bonded to the core. Consequently, the critical forces that lead to wrinkling of the face layers depend on the stiffness parameters of both the face layers and the core, the geometry of the structure, and the bonding and loading conditions.

Shouldn't it be longer wavelength?

Numerous studies have investigated this buckling mode, with each author adopting different assumptions to model the behavior of the core. One of the earliest and simplest models is the Winkler elastic foundation model, where the core's elastic properties are represented as a series of parallel springs. This model primarily addresses the core's transverse stiffness and does not account for shear interaction at the interface between the facesheet and the core. Allen [28] used a similar concept, treating the core as an elastic medium and applying Airy's stress function. However, the drawback of such models is that the stress function must be sufficiently comprehensive to capture all the interactions between the components [35].

Another common approach is to consider the core as a thick medium. Hoff and Mautner [36] adopted this method, assuming that the deformation of the core decays linearly through its thickness. Plantema [37], on the other hand, treated the core as an exponentially thick medium, with deformations decaying exponentially. Eq. (13) represents a typical form of classical wrinkling formulas. Some of these formulas are summarized in Table 3. The constant C varies between 0.4 and 1, depending on the assumptions and the specific study. The experimental study by Hoff and Mautner on sandwich structures with isotropic cores recommended a factor of 0.5. This value is widely used in sandwich design because of its conservative nature, incorporating a safety margin.

$$P_{cr} = C \sqrt[3]{E^f E_z^c G^c} \quad (13)$$

Table 3 Summary of standard wrinkling formulas

Author	Formula (in N/mm)
Allen [28]	$f B_1 E^f \frac{1}{3} E_z^c \frac{2}{3}$ where $B_1 = 3 [12(3 - \nu^c)^2(1 + \nu^c)^2]^{-\frac{1}{3}}$
Hoff and Mautner [36]	$0.5 f \sqrt[3]{E^f E_z^c G_{xz}^c}$
Plantema [37]	$1.7 f \sqrt[3]{E^f E_z^c G_{xz}^c}$
Niu and Talreja [38]	$f \left[\left(\frac{3C}{2} \right)^{\frac{2}{3}} (E^f)^{\frac{1}{3}} + (1 - \nu^c)C + C^{\frac{4}{3}} \left(\frac{3}{2E^f} \right)^{\frac{1}{3}} \right]$ where $C = \frac{E^c}{(1 + \nu^c)(3 - \nu^c)}$

In recent years, the interaction between local and global buckling—commonly referred to as mode interaction or interactive buckling—has gained significant attention, particularly in thin-walled and slender sandwich structures. This phenomenon can lead to a reduction in critical load, increased imperfection sensitivity, and unstable post-buckling behavior [39–41]. Although extensively analyzed in the context of I-beams and metallic struts, its implications for lattice-core sandwich columns remain an open topic.

To better understand such buckling phenomena, researchers have developed advanced models to capture the interaction between global buckling and local wrinkling in sandwich structures [35,38]. These models typically express the critical load as a function of the buckling wavelength and predict the onset of either global or local instability. One of the earliest unified solutions for sandwich plates with isotropic face sheets was proposed by Benson and Mayers [42], and later extended by Hadi and Matthews [43]. Frostig further expanded these models to include laminated face sheets and introduced additional refinements to enhance their predictive capabilities [44,45].

In order to improve kinematic accuracy, researchers have also begun employing higher-order functions to describe the kinematics of the sandwich column. The higher-order sandwich panel theory (HSAPT) [44] applies thin plate kinematics to the face layers and linear continuum mechanics to the core. However, a limitation of HSAPT arises in neglecting the axial stiffness of the core. While it accounts for transverse and shear stiffness, it assumes a constant shear stress distribution in the core and uses this assumption as one of the theory's generalized coordinates. Phan addressed this limitation [46] by extending the higher-order sandwich panel theory (EHSAPT). The EHSAPT incorporates axial, transverse, and shear stiffness of the core and introduces the rotation at the core centroid as an additional generalized coordinate.

An important contribution to this field is an analytical model by Vonach [47], which provides a single explicit equation for determining the critical wrinkling load of sandwich plates with isotropic face layers and thick orthotropic cores. In a parametric study, Vonach demonstrated that wrinkling behavior is highly sensitive to the in-plane stiffness of the core, particularly in structures with highly orthotropic cores (e.g., honeycomb, strut-based cores). The study's findings showed strong agreement with numerical solutions for cases involving very thick cores, where the core thickness approaches infinity and face sheet interactions become negligible. This result also highlights a key distinction between the EHSAPT and the original HSAPT.

In this context, several higher-order theories have been introduced to improve the kinematic description of sandwich structures. HSAPT accounts for transverse and shear effects of the core, while EHSAPT extends this framework by including axial stiffness and centroid rotation. More generally, Carrera [48] introduced the Carrera Unified Formulation (CUF), a hierarchical approach capable of generating a wide range of higher-order models, although its application is mathematically involved and usually requires numerical implementation. In contrast, the present closed-form higher-order formulation is restricted to beam-level analyses, yet it offers a simple and efficient alternative that captures the essential kinematic features and predicts wrinkling loads in close agreement with FEM.

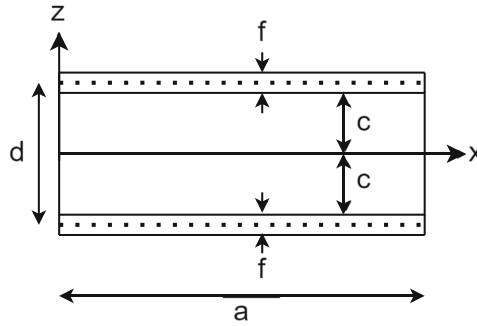


Fig. 5 Sandwich construction

4 New higher-order solution for symmetrical wrinkling

4.1 Assumptions

In this section, we present an energy-based approximate solution for a new higher-order theory. Unlike traditional approaches that assume a constant transverse displacement in the core, our model considers a third-order variation of the transverse displacement and a fourth-order variation of the axial displacement with respect to z . This formulation incorporates transverse compressibility and shear deformation of the core into the constitutive law, enhancing the predictive capabilities of the model. A closed-form linear analysis is conducted using the Ritz method, providing an explicit solution for the critical wrinkling load. The formulation aims to obtain a solution for symmetrical wrinkling in the case of a sandwich column with FBCC unit cells.

In the development of the new higher-order theory, the major assumptions taken are as follows:

1. The face sheets satisfy the Kirchhoff–Love assumption, and their thicknesses are small compared with the overall thickness of the sandwich section.
2. The core is compressible in the transverse and axial directions.
3. The core displacements must satisfy the following symmetric deformation criteria:

$$u^c(z > 0) = u^c(z < 0) \quad (14a)$$

$$w^c(z > 0) = -w^c(z < 0) \quad (14b)$$

4. The face sheets and the core are assumed to be perfectly bonded.
5. Taking into account the symmetry of wrinkling mode, only the structure lying above the mid-plane is considered for calculating the wrinkling load.

It should be noted that the present analysis framework was developed for symmetrical wrinkling, which is the mode observed in the considered sandwich columns. To extend the model toward antisymmetrical wrinkling, the displacement assumptions in Eqs. (17a) and (17b) need to be modified by enforcing $w^c(z > 0) = w^c(z < 0)$ and $u^c(z > 0) = -u^c(z < 0)$. To extend the model toward antisymmetrical wrinkling, the displacement assumptions in Eqs. (17a) and (17b) need to be modified by enforcing $u^c(z > 0) = -u^c(z < 0)$ and $w^c(z > 0) = w^c(z < 0)$. This can be achieved by representing $u^*(x, z)$ in Eq. (17a) with odd powers of z and $w^*(x, z)$ in Eq. (17b) with even powers of z . However, since antisymmetrical wrinkling was not observed in the present study, such an extension was not required.

Here, again repeated text.

4.2 Kinematic equations

Figure 5 shows a sandwich column of length a with a core of thickness $2c$ and face sheet thickness f . A cartesian coordinate system is defined at one end of the column, and its origin is placed in the middle of the core. Only loading in the $x - z$ plane is considered to act on the column which solely causes displacements in the x and z directions designated by u and w respectively. The subscript 0 refers to the middle surface of the corresponding phase. It should also be noted that the rigidities and all applied loadings are per unit width.

The face sheets are assumed to satisfy the Kirchhoff–Love assumptions and their thicknesses are small compared to the overall thickness of the sandwich section. Therefore, the displacement field for the top face sheet, $c \leq z \leq c + f$, is

$$u^t(x, z) = u_0^t(x) - \left(z - c - \frac{f}{2}\right) w_{0,x}^t(x) \quad (15a)$$

$$w^t(x, z) = w_0^t(x) \quad (15b)$$

The only non-zero strain in the face sheet is the axial strain, which is written as

$$\epsilon_{xx}^t(x, z) = u_{,x}^t(x, z) = u_{0,x}^t(x) + \left(c + \frac{f}{2} - z\right) w_{0,xx}^t(x) \quad (16)$$

In practical applications, the core of a sandwich structure can undergo transverse deformation when subjected to loading. While this deformation is often neglected as a first-order approximation in the classical sandwich model, there are many cases where its inclusion becomes essential. For instance, transverse deformation plays a significant role in the energy absorption capabilities of structures under extreme conditions, such as blast loading [49], or during wrinkling when the core material is highly compliant. To account for the transverse compressibility of the core, a higher-order series expansion in the transverse coordinate can be employed to represent both in-plane and out-of-plane displacements. In the present study, the core displacement is expressed as:

$$u^c(x, z) = \left(1 - \beta_2 \frac{z^2}{c^2}\right) u_0^c(x) + w_{,x}^c(x) \frac{z}{c} \quad (17a)$$

$$w^c(x, z) = w_0^c(x) \frac{z}{c} + \beta_3 \frac{z^3}{c^3} \quad (17b)$$

In these equations, w_0^c and u_0^c are the transverse and in-plane displacement of the middle plane of the core, while β_i ($i = 2$ and 3) are unknown functions that are to be determined from displacement continuity, as follows:

For the top face sheet/core interface, $z = c$,

$$u^c(x, z)|_{(z=c)} = u^t(x, z)|_{(z=c)} \quad (18a)$$

$$w^c(x, z)|_{(z=c)} = w^t(x, z)|_{(z=c)} \quad (18b)$$

Substitution of Eqs. (17) into the two continuity Eq. (18) leads to:

$$\beta_2 = \frac{2u_0^c(x) - 2u_0^t(x) - f w_{0,x}^t(x) + 2w_{0,x}^c(x)}{2u_0^c(x)} \quad (19a)$$

$$\beta_3 = w_0^c(x) - w_0^t(x) \quad (19b)$$

The in-plane displacement in the core is (fourth order in z):

$$u^c(x, z) = u_0^c(x) + \frac{z^2}{c^2} \left(u_0^t(x) - u_0^c(x) + \frac{f w_{0,x}^t(x)}{2} \right) - \frac{z^4}{c^4} \left(w_{0,x}^c(x) - w_{0,x}^t(x) \right) \quad (20a)$$

Thus, the transverse displacement in the core in this new higher-order core theory can be expressed as follows (third order in z):

$$w^c(x, z) = \frac{z w_0^c(x)}{c} - \frac{z^3}{c^3} \left(w_0^c(x) - w_0^t(x) \right) \quad (20b)$$

Therefore, this theory has four generalized coordinates: two for the top face sheet, w_0^t , u_0^t , and two for the core, w_0^c , u_0^c . The strains can be obtained from the displacements using linear strain–displacement relations. This leads to:

$$\epsilon_{xx}^c(x, z) = u_{0,x}^c(x) + \frac{z^2}{c^2} \left(u_{0,x}^t(x) - u_{0,x}^c(x) + \frac{f w_{0,xx}^t(x)}{2} \right) - \frac{z^4}{c^4} \left(w_{0,xx}^c(x) - w_{0,xx}^t(x) \right) \quad (21a)$$

$$\epsilon_{zz}^c(x, z) = \frac{w_0^c(x)}{c} - \frac{3z^2}{c^3} (w_0^c(x) - w_0^t(x)) \quad (21b)$$

$$\gamma_{zx}^c(x, z) = \frac{z}{c} w_{0,x}^c(x) - \frac{5z^3}{c^4} (w_{0,x}^c(x) - w_{0,x}^t(x)) + \frac{2z}{c^2} \left(u_0^t(x) - u_0^c(x) + \frac{f w_{0,x}^t(x)}{2} \right) \quad (21c)$$

4.3 Constitutive relations

The equations developed so far can be applied to general materials. In the following sections, we shall assume that the face sheets are isotropic and that the core is transversely isotropic. We use the notations $1 \equiv x$, $3 \equiv z$ and $55 \equiv zx$. The general stress–strain relationship for the face sheets is as follows:

$$\sigma_{xx}^t = C_{11}^t \epsilon_{xx}^t \quad (22a)$$

$$\sigma_{zz}^t = C_{13}^t \epsilon_{xx}^t \quad (22b)$$

where in terms of Young's modulus E_1^t and Poisson's ration, ν_{31}^t , the stiffness constant for column/wide panel are $C_{11}^t = E_1^t$ and $C_{13}^t = \nu_{31}^t E_1^t$. It is important to note that σ_{zz}^t does not ultimately enter the variational equation because the corresponding strain ϵ_{zz}^t is zero. The stress-strain relationship for the core reads:

$$\begin{bmatrix} \sigma_{xx}^c \\ \sigma_{zz}^c \\ \tau_{zx}^c \end{bmatrix} = \begin{bmatrix} C_{11}^c & C_{13}^c & 0 \\ C_{13}^c & C_{33}^c & 0 \\ 0 & 0 & C_{55}^c \end{bmatrix} \begin{bmatrix} \epsilon_{xx}^c \\ \epsilon_{zz}^c \\ \gamma_{zx}^c \end{bmatrix} \quad (22c)$$

The stiffness matrix with components C_{ij}^c in Eq. (22c) is the inverse of the compliance matrix, whose components a_{ij}^c are expressed in terms of the extensional and shear moduli and Poisson's ratio of the core, using the mechanical properties obtained from Table 1, as:

$$a_{11}^c = \frac{1}{E_1^c}; \quad a_{13}^c = \frac{-\nu_{zx}^c}{E_3^c}; \quad a_{31}^c = \frac{-\nu_{xz}^c}{E_1^c}; \quad a_{33}^c = \frac{1}{E_3^c}; \quad a_{55}^c = \frac{1}{2G_{zx}^c} \quad (22d)$$

4.4 Governing equations

The governing equation for this column buckling problem is derived using the principle of minimum potential energy, which is solved using the Ritz method. We first consider the total elastic potential of a column in the buckled state, which is composed of the internal potential Π_i and the potential Π_a of the applied forces:

$$\Pi = \Pi_i + \Pi_a \quad (23)$$

The strain energy or inner potential stored in the buckled column can be calculated as follows:

$$\Pi_i = \int_0^L \left[\int_c^{c+f} \sigma_{xx}^t(x, z) \epsilon_{xx}^t(x, z) dz + \int_0^c (\sigma_{xx}^c(x, z) \epsilon_{xx}^c(x, z) + \sigma_{zz}^c(x, z) \epsilon_{zz}^c(x, z) + \tau_{zx}^c(x, z) \gamma_{zx}^c(x, z)) dz \right] dx \quad (24)$$

Once the bifurcation state is reached, the applied load P will induce displacement w due to buckling deformations. We consider that concentrated compressive loads (per unit width) $F^t P$ and $F^c P$ are applied to the top face sheet and core, such that they sum up to the total applied load P . Additionally, the load distribution is configured to ensure that the core and face sheet experience the same axial strain, as expressed in Eq. (21a) (Fig. 6).

$$\epsilon_{xx}^t = \epsilon_{xx}^c \quad (25a)$$

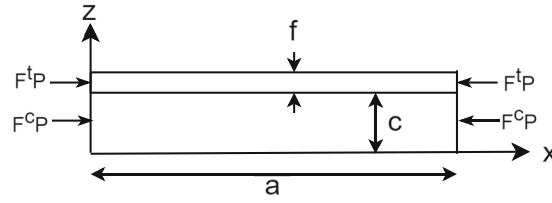


Fig. 6 Loading of sandwich column

The face sheets and the core are aligned longitudinally in parallel, ensuring consistent strain distribution across both elements. The mathematical expressions explicating this correlation are presented subsequently. Considering Hook's law $\sigma = E\epsilon$ and $\sigma = \frac{F}{A}$ and taking into consideration the isolated system's forces and interactions

$$P = F^t P + F^c P \quad (25b)$$

results in:

$$F^t = \frac{2E^f f}{E_1^c c + 2E^f f} \quad (25c)$$

$$F^c = \frac{E_1^c c}{E_1^c c + 2E^f f} \quad (25d)$$

The external potential energy, in this case, can be written as:

$$\Pi_a = -\frac{P}{2} \int_0^L \left(F^t \left(\frac{dw^t}{dx} \right)^2 + F^c \left(\frac{dw^c}{dx} \right)^2 \right) dx \quad (25e)$$

Substitution of Eqs. (25c) and (25d) into Eq. (25e) results in

$$\Pi_a = -\frac{P}{2(E_1^c c + 2E^f f)} \int_0^L \left(2E^f f \left(\frac{dw^t}{dx} \right)^2 + E_1^c c \left(\frac{dw^c}{dx} \right)^2 \right) dx \quad (25f)$$

After calculating the inner and external potential, we can substitute that expression in Eq. (23) and solve for the critical buckling load. Among the various solution techniques available, the Ritz method is often employed to obtain an approximate solution. It has been used by authors such as Hadi and Matthews [43] and Vescovini [50] to study wrinkling, as it provides a good trade-off between accuracy and computational cost. The Ritz method uses the principle of the minimum of the total elastic potential. It assumes that the total potential of the column under consideration can be represented as a function of the displacements $u_0, w_0 : \Pi = \Pi(u_0, w_0)$. The displacements are written as a linear combination of shape functions and the unknown Ritz constants. Because wrinkling is a local periodic instability, global boundary conditions have a negligible effect on critical load. FEM verification shows that the critical wrinkling load is effectively insensitive to end supports for the configurations studied. Therefore, shape functions that satisfy simply supported boundary conditions were adopted as an admissible set.

$$u_0^j(x, z) = U^j \cos \frac{n\pi x}{a} \quad (26a)$$

$$w_0^j(x, z) = W^j \sin \frac{n\pi x}{a} \quad (26b)$$

where j denotes t or c . The shape functions (26) are substituted into the total elastic potential of the column. The only quantities that are still amenable to variation are the Ritz constants U^j and W^j . As a consequence, the first variation δ of the total elastic potential transforms into the following form:

$$\delta \Pi = \frac{\partial \Pi}{\partial U^t} \delta U^t + \frac{\partial \Pi}{\partial U^c} \delta U^c + \frac{\partial \Pi}{\partial W^t} \delta W^t + \frac{\partial \Pi}{\partial W^c} \delta W^c = 0 \quad (27)$$

The variations of the constants are independent and arbitrary, so Eq. (27) holds only if:

$$\frac{\partial \Pi}{\partial U^t} = 0, \quad \frac{\partial \Pi}{\partial U^c} = 0, \quad \frac{\partial \Pi}{\partial W^t} = 0, \quad \frac{\partial \Pi}{\partial W^c} = 0 \quad (28)$$

The resulting equations are the Ritz equations, a system of homogeneous linear equations from which the constants U^j and W^j can be determined. Since our objective is to calculate the critical buckling load P , we can write these equations in matrix form as

$$\underline{K}\underline{U} = 0 \quad (29)$$

where $\underline{K}$ is a 4×4 matrix involving material stiffnesses and sandwich dimensions and is dependent on the critical buckling load P , while $\underline{U} = [U^t \ U^c \ W^t \ W^c]^T$ represents the Ritz constants. The critical load is determined by finding the value of P for which the system has a non-trivial solution. This can be done by zeroing the determinant:

$$\det |\underline{K}| = 0 \quad (30)$$

The above system of equations is an eigenvalue problem. By solving Eq. (30), we obtain eigenvalues corresponding to the buckling, and the minimum eigenvalue corresponds to the critical buckling load P .

5 Finite element method

The FEM is employed as a validation tool to assess the accuracy of the proposed solutions for sandwich columns with lattice core structures. A linear perturbation buckling analysis is carried out using the subspace solver in ABAQUS CAE on a full 3D model of the sandwich structure. The model employs 3D solid elements for the facesheets and beam elements for the lattice core. To accurately replicate the additive manufacturing process, tie constraints are used to couple the facesheets and the core, ensuring continuous face–core bonding without introducing artificial imperfections. To simulate different boundary conditions, constraints were applied directly to both face and core ends, while the concentrated axial load was applied only at the reference points. The reference points were connected to the end nodes through a kinematic coupling constraint. This setup provides a uniform transmission of axial forces and allows the application of loads while restricting unwanted cross-sectional deformations at the loaded boundaries.

Meshing is carefully tailored to the geometry and dimensions of the sandwich column to ensure mesh-independent buckling load predictions. A mesh convergence check was performed and showed that the predicted critical loads remain unchanged with reasonable refinement. Quadratic solid elements with reduced integration (C3D20R), with a mesh size of 0.5 mm, are used for the facesheets in order to avoid unnecessary computational effort. The core lattice is modeled using linear beam elements (B31) with a mesh size of 2 mm. Depending on the facesheet thickness, between 6 and 12 solid elements are used through its thickness to capture local buckling behavior accurately. A sample meshed section with a 20 mm core and 3 mm facesheet is shown in Fig. 7. This FEM strategy enables high-accuracy modeling of geometric complexity and anisotropic material behavior, captures both global and local buckling phenomena, and supports parametric studies across various core and facesheet dimensions. The numerical results are compared with analytical predictions to assess the validity of the proposed models. The four boundary conditions investigated in this study include fixed–fixed, fixed–pinned, pinned–pinned, and fixed–free.

6 Results

The complex FBCC lattice architecture investigated in this study is particularly well-suited for additive manufacturing techniques, which allow for precise control over critical geometric parameters such as strut diameter, unit cell size, and the continuity between the face sheets and the core. This manufacturing approach enables the formation of a smooth and defect-free face–core interface, thereby eliminating imperfections typically associated with bonded constructions and permitting an accurate representation of all relevant buckling modes in numerical simulations.

This section presents the results of both global and local buckling analyses for sandwich column configurations. To better illustrate the proposed methodology, the discussion begins with the wrinkling behavior,

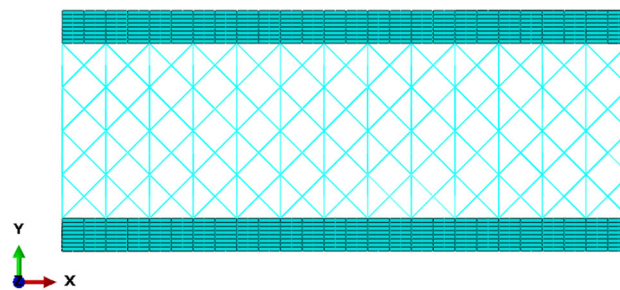


Fig. 7 A section of meshed Sandwich column

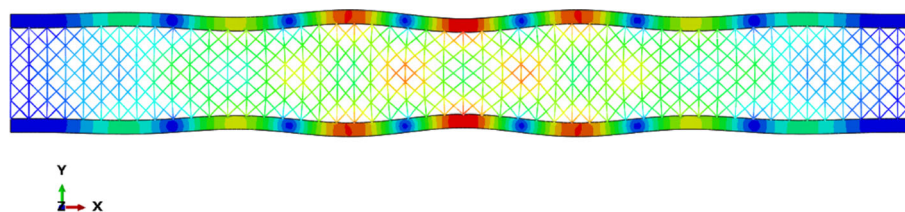


Fig. 8 Wrinkling in the sandwich column under fixed-fixed boundary conditions

followed by the development and validation of a comprehensive analytical model capable of predicting the critical loads associated with all relevant failure modes, including wrinkling. The critical buckling loads obtained from this analytical model are subsequently compared against results derived from finite element analysis (FEA). The study spans a range of core thicknesses from 8 to 24 mm in 4 mm increments, and face sheet thicknesses from 0.25 to 4 mm in 0.25 mm increments. Throughout the analysis, the unit cell size is fixed at 4 mm, and the strut diameter is maintained at 0.4 mm. Initially, the newly developed higher-order wrinkling solution is compared with FEA results. Subsequently, the complete analytical framework is evaluated through comparison with global and local buckling predictions under various boundary conditions.

6.1 Comparison of wrinkling higher-order analytical solution

Wrinkling was observed in the sandwich column under both fixed–fixed and fixed–pinned boundary conditions. In particular, the critical wrinkling load obtained from the FEM analysis remained unchanged under different boundary conditions. This observation indicates that the onset of wrinkling is governed primarily by the intrinsic characteristics of the sandwich configuration—such as core and facesheet properties—rather than by the applied boundary constraints. Figure 8 illustrates the wrinkling pattern in a sandwich column with a core thickness of 20 mm and a facesheet thickness of 2 mm under fixed–fixed boundary conditions.

Figure 9 compares the wrinkling loads predicted by the proposed higher-order analytical model with those obtained from FEM simulations under fixed–fixed boundary conditions. The results show a strong agreement between the two approaches. As the face sheet thickness increases, the wrinkling load from FEM is observed to rise in a nonlinear manner. This trend is likely due to the compliance of the FBCC core, which lacks struts oriented perpendicular to the face sheet. As a result, the core exhibits reduced transverse stiffness, leading to increased deformation. The proposed analytical model accounts for this effect through the incorporation of a higher-order displacement function, thereby enabling a more accurate representation of the core's mechanical response.

Further analysis at higher core thickness shows a slight divergence between the FEM and analytical results with increasing face sheet thickness. Nevertheless, the percentage error remains relatively low and within acceptable limits for the purposes of this study. It is also important to note that at a face sheet thickness of 0.75 mm—especially for core thicknesses of 20 mm and 24 mm—the deviation between the FEM and analytical predictions is most distinct. This may be attributed to the fact that this thickness range represents a transition regime between intracellular buckling, typically observed for thinner face sheets (e.g., 0.5 mm), and the appearance of dominant wrinkling behavior. Consequently, the wrinkling response in this intermediate zone may not fully conform to the assumptions underpinning either buckling mode.

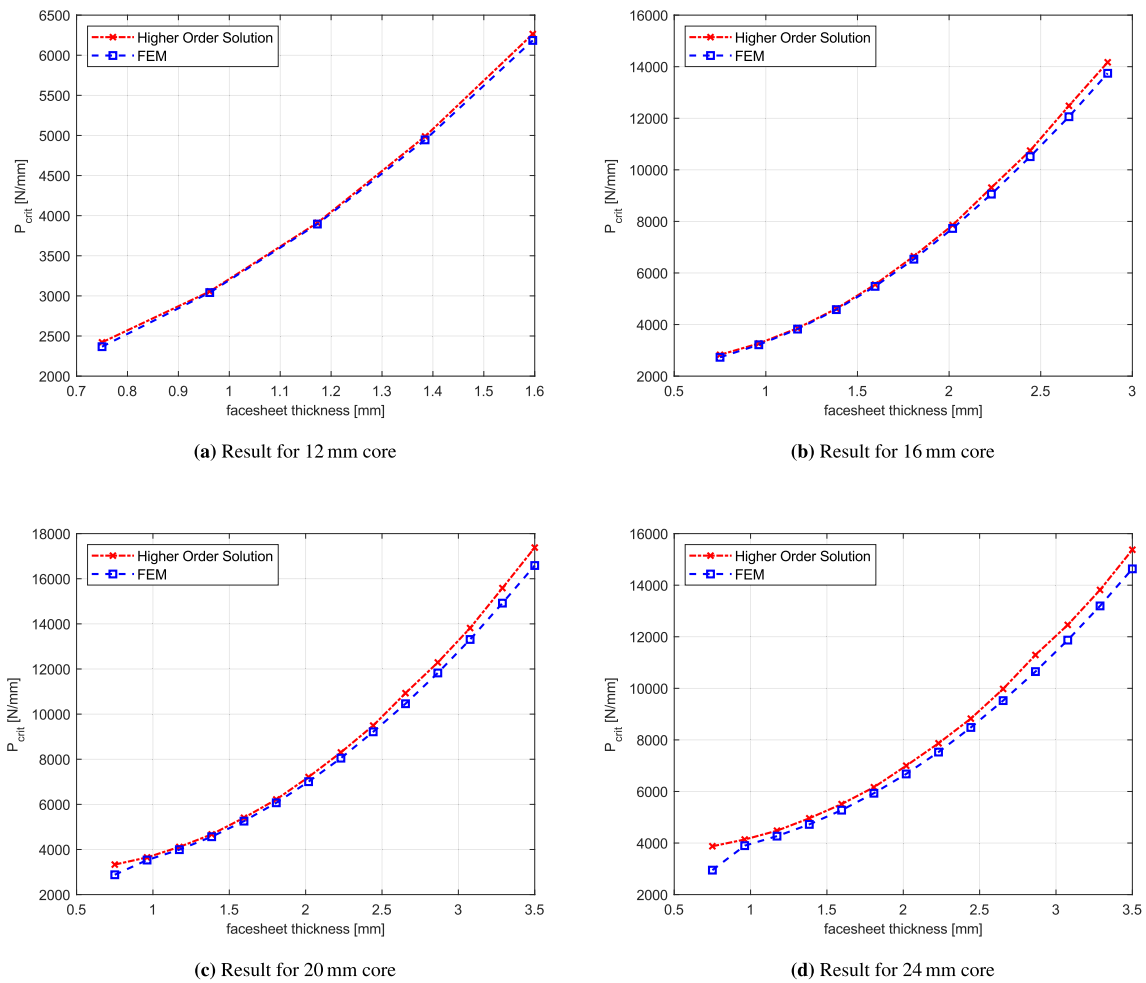


Fig. 9 Comparison of wrinkling loads predicted by the proposed higher-order analytical model and FEM results for various core thicknesses under fixed-fixed boundary condition

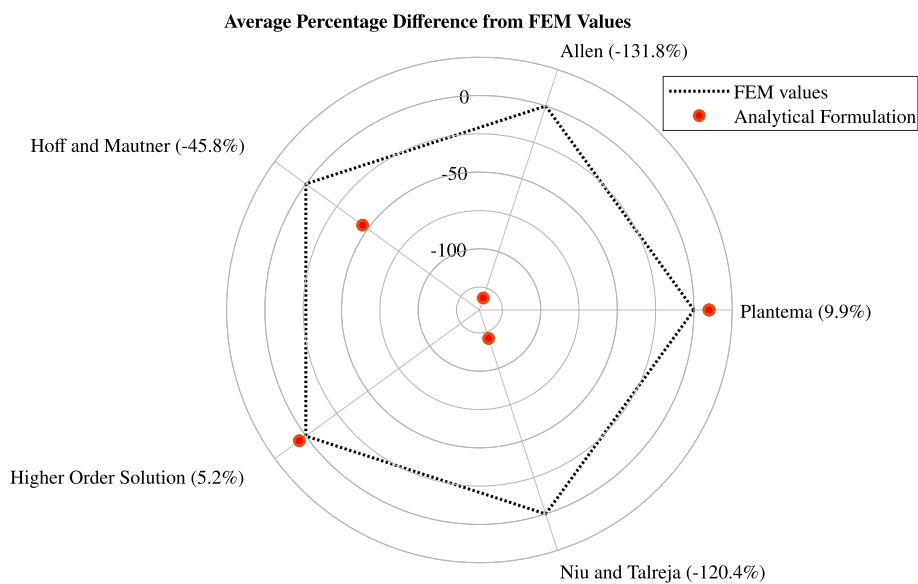


Fig. 10 Radar-type comparison between different analytical models and FEM results

6.2 Comparison between different analytical formulations for wrinkling

In order to perform a comparative assessment of the various analytical formulations listed in Table 3, alongside the higher-order solution and the FEM results, a radar-type comparison graph has been used as shown in Fig. 10. This graphical representation illustrates the relative percentage deviation between the buckling load values predicted by each analytical model and the corresponding FEM results. The degree of agreement between each formulation and the FEM reference is thereby visualized, with a perfect correlation being represented by the dotted reference line.

For a sandwich column with FBCC lattice core, it is observed that none of the models mentioned in Table 3 yield results that align satisfactorily with the expected behavior of the structure. A primary reason for this deviation can be due to the inherent assumption made in all of these models that the core behaves isotropically. However, this assumption does not hold for the current configuration, where the core is composed of a periodic FBCC unit cell structure that exhibits transverse isotropy. Among the analytical formulations considered, those proposed by Allen and by Niu & Talreja show the most pronounced deviations of approximately 130% and 120%, respectively. In analogy to Allen's model, the approach by Niu & Talreja also adopts an antiplane core representation. This modeling framework is characterized by the assumptions that the shear modulus in directions normal to the faces of the sandwich column is finite, the Young's modulus in these directions is infinite, and the Young's modulus in the in-plane directions parallel to the sandwich faces is effectively zero. As a result of these simplifications, the models incorporate only the elastic modulus in the through-thickness direction (E_z) and the Poisson's ratio (ν) of the core material. The shear modulus (G) is estimated using the classical Lamé expression, $G = \frac{E}{2(1+\nu)}$, which inherently assumes isotropic material behavior. In the current context, where the FBCC-based core exhibits transverse isotropy, this simplification leads to a significant underestimation of the core's actual stiffness and, consequently, to highly conservative predictions of the wrinkling load.

In contrast to Hoff's model, which employs a theoretical coefficient of 0.91, the use of a reduced practical coefficient of 0.5 yields results that are quite conservative. This adjustment was proposed by the original authors based on experimental observations, which may have been influenced by factors such as boundary conditions, manufacturing defects, or other inherent imperfections. It is important to note, however, that the FEM results used in the current study are based on the assumption of an idealized, defect-free structural configuration. Among the considered analytical approaches, the formulation proposed by Plantema shows better agreement with the FEM results. The principal distinction between the assumptions underlying the models by Hoff & Mautner and Plantema lies in the representation of the through-thickness stress decay: while the former adopts a linear decay, Plantema introduces an exponential decay function, similar to those commonly used in shear lag analyses. Nevertheless, even this formulation exhibits an average deviation of approximately 10% when compared to the FEM predictions. A plausible explanation for this deviation is the neglect of the axial stiffness (E_x) of the core, which, as highlighted by Vonach [47], becomes particularly significant when the core material is not isotropic. This, combined with the implementation of a higher-order displacement field in the current work, may account for the improved performance of the newly proposed solution in capturing the wrinkling behavior with greater accuracy.

6.3 Comparison of the comprehensive analytical model

The integration of a novel analytical formulation for wrinkling with established theories of global and intracell buckling has led to the development of a comprehensive analytical model. This model evaluates the critical load for all potential failure modes and identifies the dominant mode that corresponds to the minimum critical load. Specifically, Allen's formula is used for global buckling, Zenkert's formula is applied for intracell buckling, and the proposed analytical model is incorporated to account for wrinkling.

Once the formulation is established, it enables the calculation of the critical buckling load for a given sandwich structure configuration. Additionally, it determines the failure mode associated with the lowest critical load among the following types: out-of-plane buckling (P_O), in-plane buckling (P_I), intracell buckling (P_D), and wrinkling (P_W). The critical load for the structure is given by the following:

$$P \leq \min [P_O, P_I, P_D, P_W] \quad (31)$$

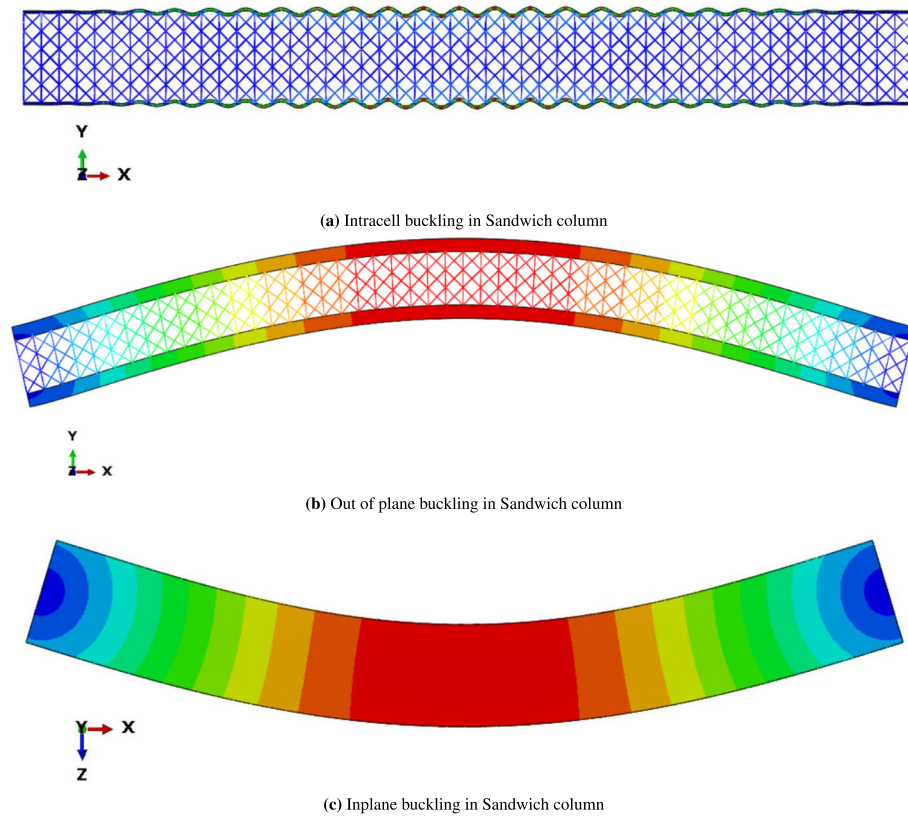


Fig. 11 Comparison of buckling modes in sandwich columns

Figure 12 presents a comparative analysis between the proposed analytical model and FEM results under various boundary conditions. For each case, a representative core thickness was selected to capture the range of buckling modes observed. This comparison demonstrates that the analytical model not only accurately predicts the critical buckling loads but also effectively captures the dominant buckling mode under each condition.

Intracell buckling, illustrated in Fig. 11a, was observed only under boundary conditions other than fixed-free. Specifically, it occurred for facesheet thicknesses of 0.25, mm and 0.5, mm, as seen in Fig. 12a, c, d. This indicates that intracell buckling is confined to cases where the facesheet is sufficiently thin. FEM simulations further reveal the presence of both symmetrical and antisymmetrical intracell buckling modes. Symmetrical buckling occurs when the number of lattice layers in the thickness direction is even, whereas antisymmetrical buckling arises when the number of layers is odd. Notably, existing literature does not distinguish between these two forms of intracell buckling. Despite this, the analytical model—based on Zenkert’s formulation—offers a quantitative prediction of intracell buckling, demonstrating the model’s ability to accurately capture these local buckling behaviors.

Global buckling was observed in the form of out-of-plane and in-plane buckling as shown in Fig. 11b, c. In-plane buckling primarily occurs when the stiffness about the z -axis is weaker than that of the y -axis. However, stiffness is not the sole governing parameter. In the current configuration, in-plane buckling was observed only under fixed-free and pinned-pinned boundary conditions. In some instances like Fig. 12c, the buckling mode transitioned from in-plane buckling to wrinkling. This observation highlights that both the stiffness characteristics of the sandwich column and the boundary conditions significantly influence the buckling mode. While it is commonly assumed that buckling is more likely to occur along the axis with weaker stiffness, the results indicate that this assumption does not always hold.

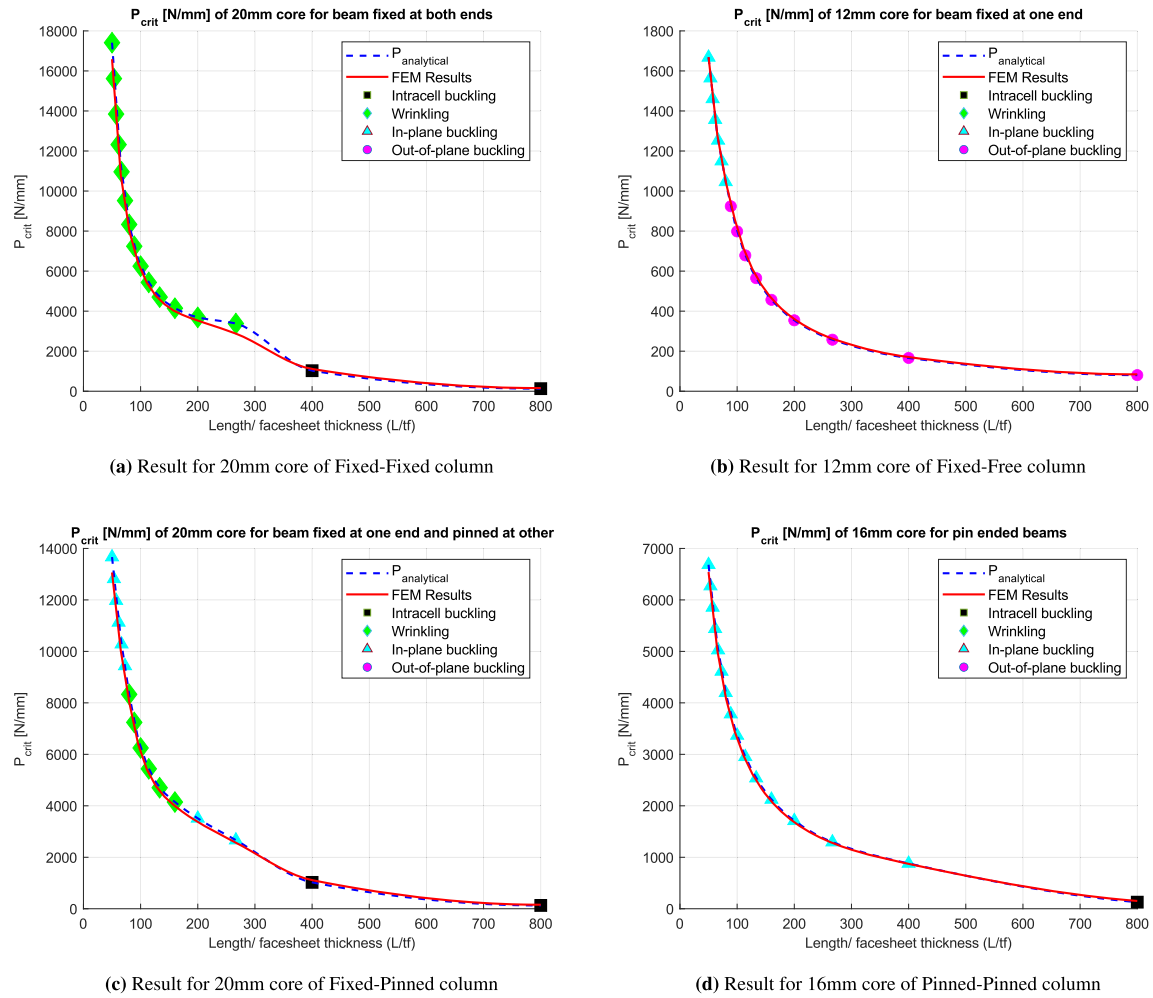


Fig. 12 Comparison of final analytical solution for FBCC with FEM results for each boundary condition

7 Conclusion

This study investigates both global and local buckling of sandwich columns with FBCC unit cells. While global buckling and intracell buckling are evaluated using established formulations from the literature, a novel higher-order approximate analytical solution is developed for the wrinkling mode, which constitutes the main contribution of this work. Global buckling of the sandwich column is considered in the form of out-of-plane and in-plane buckling. The primary distinction between these two modes lies in the plane of deformation, which is governed by the stiffness characteristics of the column and the boundary conditions. The column buckles in the direction where the stiffness is weaker. Allen's formula was employed for global buckling analysis because it effectively distinguishes between thin and thick cores and incorporates the bending stiffness of the sandwich column accordingly.

Local buckling of facesheet of the sandwich column was studied for the case of intracell buckling and wrinkling. While no dedicated formulation exists for a lattice unit cell, Zenkert's formula for corrugated cores provided a quantitative estimate for intracell buckling. For wrinkling, a novel higher-order approximate analytical solution was developed for the symmetric mode. This solution incorporates a third-order variation of transverse displacement and a fourth-order variation of axial displacement with respect to the transverse coordinate z , allowing for a more accurate representation of the wrinkling behavior.

Finite element analysis was conducted to verify the proposed analytical model, showing strong agreement between numerical and analytical results. The analysis was performed for a range of core and face sheet thicknesses, and the model successfully captured the transition between local and global buckling modes. The

results indicate that both the dimensions of the sandwich structure and the boundary conditions play critical roles in determining the buckling mode. However, the study also revealed that boundary conditions do not influence the critical load value for local buckling, suggesting that an analytical solution for all boundary conditions is unnecessary when estimating local buckling loads.

Given the efficiency of the proposed implementation, it is also interesting to provide insight into the computational time required for typical calculations. The time needed to compute the wrinkling load using FEM on a CPU with 16 GB of RAM, an Intel Core i5 processor at 2.81 GHz, is approximately 250 s. In contrast, the proposed higher-order analytical solution delivers equivalent results in under 6 s, making it a significantly more efficient alternative.

One limitation of the current model is the omission of local buckling within the lattice core, commonly referred to as strut buckling. Incorporating this failure mode in future work would enhance the model's applicability, particularly for configurations with slender core members prone to instability. Additionally, the analysis assumes ideal lattice geometry and perfect bonding between the core and facesheets, which may not fully capture the effects of manufacturing imperfections or material heterogeneities. Despite these assumptions, the closed-form nature of the solution offers significant advantages in terms of computational efficiency, making it well-suited for preliminary design, optimization, and parametric studies.

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Declarations

Conflict of interest The authors declare no conflict of interest.

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Appendix A. Elements of $\underline{K}$ matrix

The $\underline{K}$ matrix is 4 X 4 symmetric matrix and has the following elements, k_{ij} , $i, j = 1 \dots 4$:

$$k_{11} = \frac{C_{55}^c L(\sin(2n\pi) + 2n\pi)}{3cn\pi} - \frac{C_{11}^c cn\pi(\sin(2n\pi) - 2n\pi)}{20L} - \frac{C_{11}^t fn\pi(\sin(2n\pi) - 2n\pi)}{4L} \quad (\text{A.1})$$

$$k_{12} = -\frac{C_{55}^c L(\sin(2n\pi) + 2n\pi)}{3cn\pi} - \frac{C_{11}^c cn\pi(\sin(2n\pi) - 2n\pi)}{30L} \quad (\text{A.2})$$

$$k_{13} = \frac{3C_{13}^c(\sin(2n\pi) - 2n\pi)}{20} + \frac{C_{55}^c(\sin(2n\pi) + 2n\pi)}{10} + \frac{2C_{55}^c(\sin(2n\pi) + 2n\pi)}{5c} + \frac{C_{55}^c f(\sin(2n\pi) + 2n\pi)}{6c} \\ - \frac{C_{11}^c cn^2\pi^2(\sin(2n\pi) - 2n\pi)}{28L^2} - \frac{C_{11}^c cf n^2\pi^2(\sin(2n\pi) - 2n\pi)}{40L^2} \quad (\text{A.3})$$

$$k_{14} = \frac{C_{55}^c(\sin(2n\pi) + 2n\pi)}{15} - \frac{C_{13}^c(\sin(2n\pi) - 2n\pi)}{15} - \frac{2C_{55}^c(\sin(2n\pi) + 2n\pi)}{5c} + \frac{C_{11}^c cn^2\pi^2(\sin(2n\pi) - 2n\pi)}{28L^2} \quad (\text{A.4})$$

$$k_{22} = \frac{C_{55}^c L(\sin(2n\pi) + 2n\pi)}{3cn\pi} - \frac{2C_{11}^c cn\pi(\sin(2n\pi) - 2n\pi)}{15L} \quad (\text{A.5})$$

$$k_{23} = \frac{C_{13}^c(\sin(2n\pi) - 2n\pi)}{10} - \frac{C_{55}^c(\sin(2n\pi) + 2n\pi)}{10} - \frac{2C_{55}^c(\sin(2n\pi) + 2n\pi)}{5c} - \frac{C_{55}^c f(\sin(2n\pi) + 2n\pi)}{6c} \\ - \frac{C_{11}^c cn^2\pi^2(\sin(2n\pi) - 2n\pi)}{70L^2} - \frac{C_{11}^c cf n^2\pi^2(\sin(2n\pi) - 2n\pi)}{60L^2} \quad (\text{A.6})$$

$$k_{24} = \frac{C_{13}^c(\sin(2n\pi) - 2n\pi)}{15} - \frac{C_{55}^c(\sin(2n\pi) + 2n\pi)}{15} + \frac{2C_{55}^c(\sin(2n\pi) + 2n\pi)}{5c} + \frac{C_{11}^c n^2 \pi^2 (\sin(2n\pi) - 2n\pi)}{70L^2} \quad (A.7)$$

$$k_{33} = \frac{3C_{13}^c n \pi (\sin(2n\pi) - 2n\pi)}{28L} + \frac{(\sin(2n\pi) - 2n\pi)(180C_{13}^c n^2 \pi^2 - 756C_{33}^c L^2 + 126C_{13}^c c f n^2 \pi^2)}{1680Lcn\pi} \\ + \frac{3C_{13}^c f n \pi (\sin(2n\pi) - 2n\pi)}{40L} + \frac{C_{55}^c f n \pi (\sin(2n\pi) + 2n\pi)}{10L} - \frac{C_{11}^c n^3 \pi^3 (\sin(2n\pi)/2 - n\pi)}{18L^3} \\ - \frac{C_{11}^c f^3 n^3 \pi^3 (\sin(2n\pi) - 2n\pi)}{48L^3} - \frac{C_{11}^c c f^2 n^3 \pi^3 (\sin(2n\pi) - 2n\pi)}{80L^3} + \frac{2C_{55}^c f n \pi (\sin(2n\pi) + 2n\pi)}{5Lc} \\ - \frac{C_{11}^c c f n^3 \pi^3 (\sin(2n\pi) - 2n\pi)}{28L^3} + \frac{C_{55}^c f^2 n \pi (\sin(2n\pi) + 2n\pi)}{12Lc} + \frac{C_{55}^c n \pi (\sin(2n\pi) + 2n\pi)(c + 4)^2}{28Lc} \\ - \frac{C_{11}^c c f n^3 \pi^3 (\sin(2n\pi) - 2n\pi)}{28L^3} \quad (A.8)$$

$$k_{34} = \frac{C_{55}^c n \pi (\sin(2n\pi) + 2n\pi)}{5L} - \frac{23C_{13}^c n \pi (\sin(2n\pi) - 2n\pi)}{280L} \\ - \frac{(\sin(2n\pi) - 2n\pi)(138C_{13}^c n^2 \pi^2 - 336C_{33}^c L^2 + 28C_{13}^c c f n^2 \pi^2)}{1680Lcn\pi} + \frac{C_{55}^c cn\pi (\sin(2n\pi) + 2n\pi)}{20L} \\ - \frac{C_{13}^c f n \pi (\sin(2n\pi) - 2n\pi)}{60L} + \frac{C_{55}^c f n \pi (\sin(2n\pi) + 2n\pi)}{30L} + \frac{C_{11}^c n^3 \pi^3 (\sin(2n\pi)/2 - n\pi)}{18L^3} \\ - \frac{C_{55}^c f n \pi (\sin(2n\pi) + 2n\pi)}{5Lc} + \frac{C_{11}^c c f n^3 \pi^3 (\sin(2n\pi) - 2n\pi)}{56L^3} - \frac{C_{55}^c n \pi (\sin(2n\pi) + 2n\pi)(c + 4)^2}{28Lc} \quad (A.9)$$

$$k_{44} = \frac{2C_{13}^c n \pi (\sin(2n\pi) - 2n\pi)}{35L} - \frac{2C_{55}^c n \pi (\sin(2n\pi) + 2n\pi)}{5L} - \frac{C_{55}^c cn\pi (\sin(2n\pi) + 2n\pi)}{60L} \\ - \frac{(\sin(2n\pi) - 2n\pi)(336C_{33}^c L^2 - 96C_{13}^c n^2 \pi^2)}{1680Lcn\pi} - \frac{C_{11}^c n^3 \pi^3 (\sin(2n\pi)/2 - n\pi)}{18L^3} + \frac{C_{55}^c n \pi (\sin(2n\pi) + 2n\pi)(c + 4)^2}{28Lc} \\ - \frac{E_1^c Pcn\pi (\sin(2n\pi) + 2n\pi)}{4L(E_1^c c + 2E^f f)} \quad (A.10)$$

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